

Assignment2

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1 Probabilities (Theoretical)

a. Maximum Likelihood Estimation(MLE)

First we need the likelihood function based on the prob distribution.

$$L(\theta) = \prod_{i=1}^N f(x_i; \theta)$$

In this case we have :

$$L(\theta) = \theta^{2N} \prod_{i=1}^N (x_i + 1) \prod_{i=1}^N (1 - \theta)^{x_i}$$

To simplify it we will use the log-likelihood function

$$\ell(\theta) = 2N \log \theta + \sum_{i=1}^N \log(x_i + 1) + \sum_{i=1}^N x_i \log(1 - \theta)$$

The term

$$\sum_{i=1}^N \log(x_i + 1)$$

does not depend on θ , so we can ignore it when differentiating. Therefore :

$$\ell(\theta) = 2N \log \theta + \sum_{i=1}^N x_i \log(1 - \theta)$$

Now, we differentiate the log-likelihood function with respect to θ to find the value of θ that maximizes the likelihood.

$$\frac{d\ell(\theta)}{d\theta} = \frac{2N}{\theta} - (1-\theta) \sum_{i=1}^N x_i$$

Set the derivative equal to zero to find the critical points . Thus, the maximum likelihood estimate (MLE) for θ is:

$$\hat{\theta} = \frac{2N}{2N + \sum_{i=1}^N x_i} \quad (1)$$

Now if the second derivative is negative, the function is concave (curving downward), and a critical point is a maximum.

$$\frac{d^2\ell(\theta)}{d\theta^2} = -\frac{2N}{\theta^2} - \frac{\sum_{i=1}^N x_i}{(1-\theta)^2}$$

Since both terms are negative for $\theta \in [0, 1]$, the second derivative is negative, confirming that the log-likelihood is concave and that $\hat{\theta}$ is indeed a maximum.

b. Calculate $\hat{\theta}$

Now, let's apply the given samples:

$$x = [3.2, 1.4, 2.2, 7, 0.5, 3.3, 9, 0.15, 2, 3.21, 6.13, 5.5, 1.8, 1.2, 11]$$

in equation 1

The number of samples is $N = 15$. So the sum of the sample values is:

$$\sum_{i=1}^{15} x_i = 57.69$$

Substitute the values into the equation 1:

$$\hat{\theta} \approx 0.342$$

Thus, the value of $\hat{\theta}$ is approximately 0.342.

2 Naive Bayes (Theoretical)

a. $\mathbb{P}(U = 0 \mid x = 0, y = 0, z = 1)$

We want to compute the probability:

$$\mathbb{P}(U = 0 \mid x = 0, y = 0, z = 1)$$

Using Bayes' Theorem:

$$\mathbb{P}(U = 0 \mid x = 0, y = 0, z = 1) = \frac{\mathbb{P}(U = 0) \cdot \mathbb{P}(x = 0 \mid U = 0) \cdot \mathbb{P}(y = 0 \mid U = 0) \cdot \mathbb{P}(z = 1 \mid U = 0)}{\mathbb{P}(x = 0, y = 0, z = 1)} \quad (2)$$

Now, we will calculate each term individually.

- $\mathbb{P}(U = 0) = \frac{\text{Number of samples where } U=0}{\text{Total number of samples}} = \frac{3}{7}$
- $\mathbb{P}(x = 0 \mid U = 0) = \frac{2}{3}$
- $\mathbb{P}(y = 0 \mid U = 0) = \frac{1}{3}$
- $\mathbb{P}(z = 1 \mid U = 0) = 1$
- $\mathbb{P}(x = 0, y = 0, z = 1) = \frac{2}{7}$

Putting It All Together. We substitute these values into Bayes' theorem 2 and finally have :

$$\mathbb{P}(U = 0 \mid x = 0, y = 0, z = 1) = \frac{1}{3}$$

b. Why Naive Bayes fails ?

Laplace correction is used to avoid issues when one or more probabilities in the Naive Bayes formula become zero due to unseen or missing combinations of feature values in the training dataset. Without smoothing, a zero probability for any feature would cause the entire posterior probability to become zero. In question a. we did not use such a sample.

In case we want to find $\mathbb{P}(U = 0 \mid x = 0, y = 0, z = 0)$ we will need to exploit Laplace correction because $\mathbb{P}(z = 0 \mid U = 0) = 0$.

c. $\mathbb{P}(U = 1 \mid x = 0)$

We want to compute the probability:

$$\mathbb{P}(U = 1 \mid x = 0)$$

Using Bayes' Theorem:

$$\mathbb{P}(U = 1 \mid x = 0) = \frac{\mathbb{P}(U = 1) \cdot \mathbb{P}(x = 0 \mid U = 1)}{\mathbb{P}(x = 0)} \quad (3)$$

- $\mathbb{P}(U = 1) = \frac{4}{7}$
- $\mathbb{P}(x = 0 \mid U = 1) = \frac{1}{2}$
- $\mathbb{P}(x = 0) = \frac{4}{7}$

Thus using 3, the predicted probability $P(U = 1 \mid x = 0)$ is:

$$\mathbb{P}(U = 1 \mid x = 0) = \frac{1}{2}$$

3 Naive Bayes Classifier (Programming)

d. Hyperparameter L

L	Average categorical Accuracy (%)
0	85.37
1×10^{-12}	85.00
1×10^{-6}	86.06
0.1	86.06
0.5	85.85
1	86.01
5	85.25
10	84.18
50	81.25
100	79.18

Table 1: Average Categorical Accuracies on test for different L values over 100 illustrations

The hyperparameter L is used for Laplace smoothing in Naive Bayes, particularly when dealing with categorical data. Laplace smoothing is important for handling cases where a feature-value combination does not appear in the training set (i.e., when the count of some feature-values is zero). Without smoothing, these unseen feature-value combinations would have a probability of zero, and since Naive Bayes multiplies probabilities, any zero probability would result in the entire posterior probability being zero.

As shown in Table 1, small values of L might give the model higher accuracy if the training set is large and well-representative of the test set. Although Large L values may help avoid overfitting when the training set is small or when there are many unseen feature-value combinations, but excessive smoothing could lead to underfitting.